## 1. Experiment in short

| | |
|---|---|
| Number of nodes | 3 |
| Number of topologies | 1 |
| Location of the experiment | Morocco and Western Sahara $(30.47728°N, 10.035655°W)$ |
| Measurement duration | 7991 seconds |
| Measurement granularity | 1 second |
| Link quality measurements | RSSI |

## 2. Location

The experiment took place on November 6, 2015 along the coasts of Morocco and Western Sahara.

The location $(30.422180°N, 10.188453°W)$ is offshore, about 40 km far from the coast of Morocco, and the water depth is 127 m. The acoustic link connection is available only in line-of-sight conditions. A map illustrating the experiment setup is shown in Fig. 1. The experiment included 3 nodes, communicating using EvoLogics modems. The nodes were deployed from a working boat (Node 1) and the NATO Research Vessel Alliance (Nodes 2 and 3). Nodes 2 and 3 were placed 60 m from each other at a depth 10 m, as they were deployed from the stern and bow of the Alliance, respectively.

The experiment started at 9:00 (CET) and lasted for one our. The working boat was used. The working boat was displaced at two different distances: first, from 9:00 to 9:30 at about 600 m from the Alliance, then, from 9:32 to 10:00 at about 350 m. Three nodes were used: one at the stern, composed of an EvoLogics modem, a projector and an hydrophone, one at the bow of the Alliance, composed of an hydrophone and an EvoLogics modem, and one at the working boat, also composed of an EvoLogics modem and an hydrophone. During the experiments all the propellers, the engines and the echo-sounder were turned off. Both the working boat and the RV Alliance were drifting. At 9:43 (CET), 43 minutes after the beginning of the experiment, the working boat's engine was turned on in order to recharge the boat's batteries. This implied the presence of an additional noise source).

## 3. Topologies

The network included three nodes, which were drifting during the experiment. The network topology is fully connected. We list in the following the nodes' locations with respect to the positions marked in Fig. 1.

- Node 1 30.422054 N, 10.191462 W
- Node 2 30.425250 N, 10.187096 W
- Node 3 30.425048 N, 10.186467 W

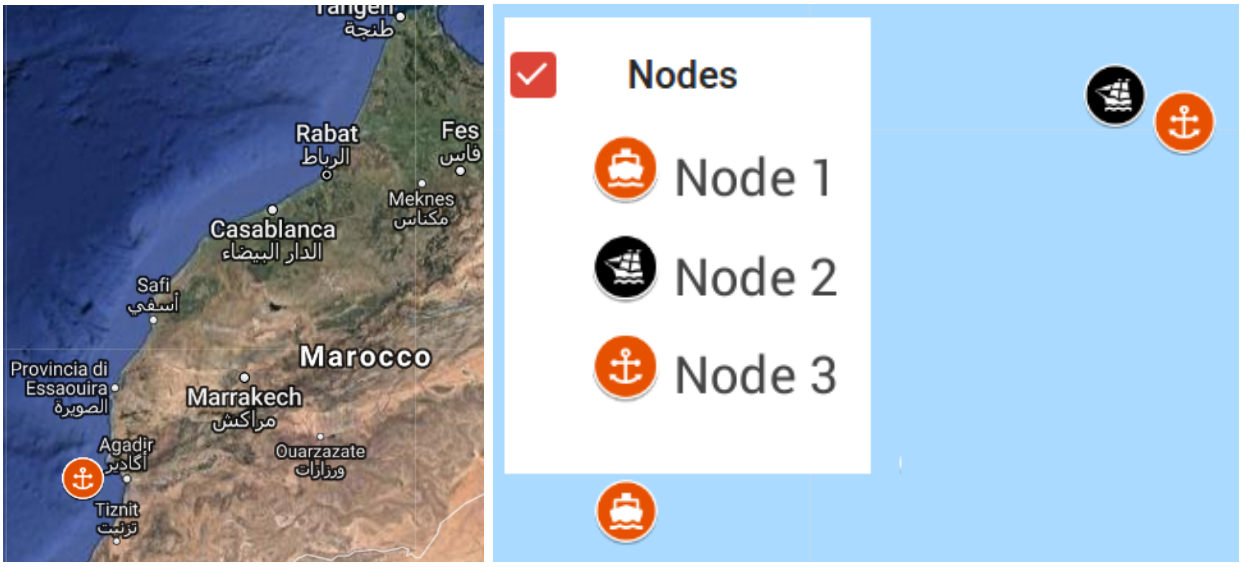



Fig. 1. Satellite picture of the sea trial location and zoom of the topology.

## 4. Description of the Nodes

The modems were EvoLogics SC2R 18/34 units, operating in the 18-34 kHz frequency band using EvoLogics's proprietary sweep-spread carrier modulation scheme. For further information, the interested reader is referred to EvoLogics's web site: https://www.evologics.de/en/products/acoustics/index.html

## 5. Tested Protocols and Schemes

The nodes were used to acquire data concerning the link quality at Node 1 in the 18-34 kHz bandwidth. In addition, a jamming test was performed, where Node 2 jammed the transmission from Node 1 to Node 3. Therefore, despite the full network connectivity no packets were transmitted between Node 1 and Node 3.

## 6. Link Quality Measurements

The network links are evaluated by means of the received signal strength indicator (RSSI). Transmissions were carried out using the EvoLogics modems' instant message mode, which allows the user to send data while excluding any further control exerted by the modem's DMAC protocol. Upon transmission, the RSSI was returned directly by the receiving modem's firmware.

## 7. Available Data

The stored data is given in the `Alomex15.mat` file. This file brings together 800 measurement epochs, provided at constant intervals of 1 second starting from the beginning of each experiment. The file contains four types of matrices:

- the location matrix `LocMat`, of size 800×3×3;
- the topology measurement matrix `TopMat`, of size 800×3×3;
- the adjacency matrix `AdjMat` of size 800×3×3;
- the technology matrix `TechMat` of size 800×3×1.

Using Matlab/Octave matrix notation:

- `LocMat(t,i,:)` are the UTM coordinates (zone 29R) and the depth of node $i$ at time instant $t$;
- `TopMat(t,i,j)` is the RSSI for link $i \rightarrow j$ at time $t$;
- `AdjMat(t,i,j)` is equal to 1 if there exists a topology link between nodes $i$ and $j$ at time $t$;
- `TechMat(t,i,p)` is equal to 1 if node $i$ can use physical layer technology $p$ at time $t$.

## 8. Reference to detailed description

Further details about this experiment are given in the following papers:

- R. Diamant, F. Campagnaro, M. De Filippo De Grazia, P. Casari, A. Testolin, V. Sanjuan Calzado, M. Zorzi, "On the Relationship between the Underwater Acoustic and Optical Channels," *IEEE Transactions on Wireless Communications*, vol. 16, no. 12, pp. 8037–8051, Dec. 2017.